Experiment 13
Stokes' Law

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Abstract

This experiment aims to measure the viscosity and drag coefficient of two different liquids, motor oil and glycerin, using Stokes' method. By applying Stokes' law, Archimedes' principle, and Newton's second law, we were able to calculate the limit velocity and determine the viscosity of the liquids. The experiments were conducted using steel and lead spheres, and the times taken for the spheres to travel a known distance in the fluids were recorded. The effects of the tube walls were accounted for using the Ladenburg correction. The results revealed the viscosity coefficients for glycerin and motor oil to be approximately 0.2139 ± 0.0089 Pa·s and 0.7308 ± 0.0076 Pa·s, respectively. Despite experimental errors, the measured values for the motor oil experiment were in reasonable agreement with theoretical predictions, whereas the results of the glycerin experience were not satisfacotry.

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1 Introduction

1.1 Contextualization

Fluids include not only liquids, but also gasses. However, in this experiment we will focus on the study of liquids.

Fluid Mechanics is the branch of physics that investigates fluids and the forces that act on them. One of the pioneers of this subject is Arquimedes, who made significant discoveries in fluids statics and formulated the equation..., nowadays known as the Archimedes' principle. Other scientifics continued his work, such as Daniel Bernoulli, Isaac Newton and Osborne Reynolds, who contributed to great advances in the understanding of viscosity and turbulence [1].

The main goal of this experiment is to measure the viscosity and drag coefficient of a liquid -in this case motor oil and glycerin- by Stokes' method, to determine its Reynold's number and to prove Stokes' law [2].

1.2 Theorical principles

1.2.1 Drag on a submerged body

When a mass goes through a fluid it experiences a force that oposses to its movement -comparable to the friction force commonly studied in Classical Mechanics-, defined as drag force. This magnitude obeys the following relation [2].

$$F_D = C_D \left(\frac{1}{2}\rho v^2\right) A \tag{1}$$

where:

- ρ is the liquid's density
- v is the velocity of the body relative to the fluid
- A is the body's area that opposes the flow
- C_D is the drag coefficient, determined by the geometric shape of the body and its orientation with respect to the flow; in addition to the fluids Reynold's number N_R, defined as:

$$N_R = \frac{\rho v D}{\eta} \tag{2}$$

where D is the diameter of the particle and η is the coefficient of viscosity of the fluid, which is measured in $Pa{\cdot}s$. The viscosity is usually measured in poises (P) too, honoring the french physicist Poiseuille [3].

1.2.2 Stokes' Law

For small values of N_R we are able to estimate the drag force on a smooth sphere, such as the ones used in the experiment. To do so, it is necessary to make use of Stokes' Law [3].

$$F_D = 3\pi\eta D v \tag{3}$$

Combining equations 1 and 2 and considering that the effective cross-section of a sphere is $A = \pi\left(\frac{D}{2}\right)^2$, it is possible to obtain a simpler definition of C_D.

$$C_D \left(\frac{1}{2}\rho v^2\right) \pi \left(\frac{D}{2}\right)^2 = 3\pi\eta D v \Rightarrow C_D \frac{1}{4}\rho v D = 3\eta \Rightarrow C_D = \frac{24\eta}{\rho v D} = \frac{24}{R} \tag{4}$$

1.2.3 Viscosity calculation

To determine the viscosity of the fluid studied we need to apply Stokes' Law (3), Archimedes Principle (10) and Newton's Second Law (11).

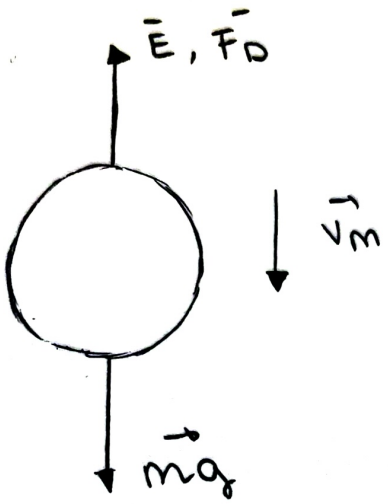


Figure 1: Forces diagram

The forces acting on the particle are the buoyancy E, the drag force F_D and the weight, defined as $P = mg$.

When the mass m reaches its limit velocity, the sum of the forces that suffers is equal to 0 as a consequence of Newton's First Law. Therefore,

$$E + F_D = mg \Rightarrow m_f g + 3\pi\eta D v = mg$$

$$3\pi\eta D v = g(m - m_f) = g(mV - m_f V_f)$$

Knowing that the V of the mass is the same as the volume V_f of the displaced fluid, we can obtain the limit velocity of the particle.

$$3\pi\eta D v = gV(m - m_f) = g\frac{4}{3}\pi\left(\frac{D}{2}\right)^3(\delta - \rho)$$

$$v_{lim} = \frac{gD^2}{18\eta}(\delta - \rho) \tag{5}$$

However, the velocity calculated in equation 5 is different from the experimental value v_m due to the influx of the tube walls. The real limit velocity can be obtained with Ladenburg correction.

$$v_{lim} = \left(1 + 2,4\frac{D}{\phi}\right)v_m \tag{6}$$

where ϕ is the inner diameter of the tube through which the experimental balls fall.

2 Materials and Methods

The materials required to carry out this experiment are the following [2].

- **Test tubes**. We used two tubes, one of inner diameter $\phi = 0,03820 \pm 0,00029m$ that contains motor oil ($\rho = 0.8913g/cm3$), and the other one of $\phi = 0,03100 \pm 0,00029m$ filled with glycerin ($\rho = 1.26g/cm3$).
- **Steel and lead spheres**. We'll need steel and lead spheres, quasi-identical. The steel ones have a diameter of $0,00200 \pm 0,00029m$ and a mass of $32,610^{-6} \pm 10^{-4}kg$, whereas the lead spheres have a diameter of $0,00270 \pm 0,00029m$ and a mass of $13910^{-6} \pm 10^{-4}kg$.
- **Chronometer**. It is a device used to measure time. We used a chronometer with a resolution of 0,01 s.
- **Thermometer**. It measures the temperature with a precission of $1C$.
- **Palmer's micrometer and ruler**. Instruments used to measure lengths with great precission. They are graduated in millimeters.

  The method we followed is quite simple. We had to drop a sphere from a small distance above the surface of the fluid, and in the center of that surface in order to avoid the effect of

the tube's walls. We measured 50 times the time it takes the sphere to descend a distance of $\Delta x = 0,22000 \pm 0,00029m$ for the steel balls and of $\Delta x_O = 0,34200 \pm 0,00029m$ for the lead balls. With this data it is possible to obtain v_m.

$$v_m = \frac{\Delta x}{\Delta t} m/s \tag{7}$$

3 Results and discussion

From all of the times measured, we have obtained an average time of $3,859 \pm 0,013$ seconds for the glycerin experiment and $2,115 \pm 0.012$ for the motor oil one. Applying the formula 7 we have computed:

- $v_m = \frac{0,220}{3,859} = 0,057009 \pm 0,000077m/s$ for the steel spheres
- $v_m = \frac{0,342}{2,1154} = 0,16167 \pm 0,00014m/s$ for the lead spheres

Nonetheless, it is necessary to take into the account the effect of the tube's walls on the limit velocity of the experimental balls. This influence can be computed with Ladenburg's correction (3).

$$v_l = \left(1 + 2,4\frac{0,002}{0,031}\right) \cdot 0,0570 = 0,0664 \pm 0,0013m/s$$

Analogously, for the motor oil experiment:

$$v_l = \left(1 + 2,4\frac{0,0027}{0,0382}\right) \cdot 0,01617 = 0,1644 \pm 0.0038m/s$$

Finally, clearing η from equation 5 we were able to determine the value of the viscosity coefficient.

$$\eta = \frac{D^2(\delta - \rho)g}{18v_l} \tag{8}$$

For the glycerin experiment we obtained a value of:

$$\eta = \frac{0,002^2 \cdot (7783 - 1260) \cdot 9,8}{18 \cdot 0,0664} = 0,2139 \pm 0,0089Pa \cdot s$$

On the other hand, the motor oil viscosity coefficient is:

$$\eta = \frac{0,0027^2 \cdot (13487 - 891.3) \cdot 9,8}{18 \cdot 0,1644} = 0,7308 \pm 0,0076Pa \cdot s$$

From equation 2 we obtain a $N_R = 0,541 \pm 0,091$ for the motor oil and a $N_R = 0,723 \pm 0,024$ for the glycerin. Finally, we obtain values of the drag coefficent of $44,330 \pm 0,072$ and $33,190 \pm 0,012$ for the motor oil and glycerin respectively.

To conclude the results and discussion section, two histograms have been plotted. 2 refers to the motor oil experience meanwhile 3 relates to the glycerin one.

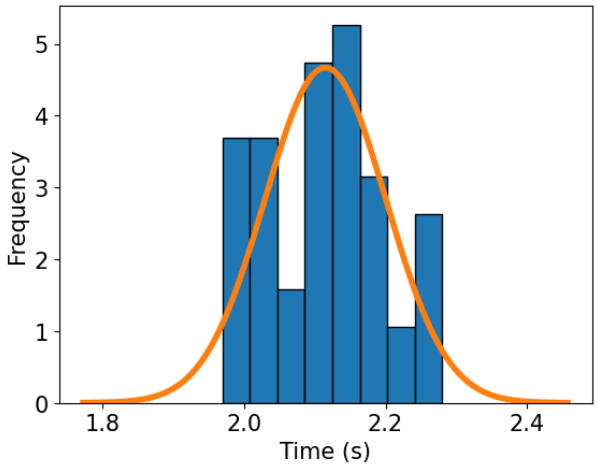


Figure 2: Histogram of frequency *vs* time that the experimental spheres take to descend a distance Δx m in the motor oil experiment.

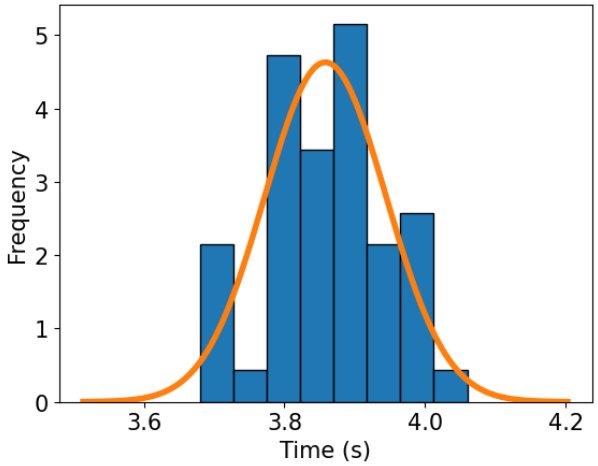


Figure 3: Histogram of frequency *vs* time that the experimental spheres take to descend a distance Δx m in the glycerin experiment.

4 Conclusions

It is very likely that errors occurred during the experiment, which may have affected the measurements and results. For instance, the value of the glycerin viscosity coefficient we obtained is $0,2139 \pm 0,0089 Pa \cdot s$, which is far away from the real value 1,410 Pa·s (for $T = 20C$) [3].

We thought this difference was because the temperature we measured in the laboratory was of $23C$, and the viscosity depends on this magnitude as established by William Sutherland [4].

$$\eta = \frac{\eta_0 \cdot (T/T_0)^{\frac{3}{2}} \cdot (T_0 - S)}{T + S} \tag{9}$$

where S is the Sutherland constant. With this equation we obtained a value of 1,49 Pa·S, which is even more different from the one we obtained. From these mistake the rest of the calculations are also different from the reality.

On the other hand, the value of the motor oil viscosity coefficient we calculated is of $0,7308 \pm 0,0076 Pa \cdot s$, which is quite close to the real one (0,6-0,8 Pa·s)[5].

Therefore, we conclude that some of the measurements of the glycerin may be wrong, or maybe we have made a mistake in our calculations. Nonetheless, the experiment was conducted

smoothly and without issues, allowing all data to be collected correctly and without complications during the process.

5 Appendixes

5.1 Elemental equations

$$E = m_f g \tag{10}$$

where m_f refers to the mass of the displaced fluid.

$$F = ma \tag{11}$$

5.2 Calculation of Uncertainties

5.2.1 Uncertainty of t

In the first place, we need to obtain the standard deviation $\sigma(t)$.

$$\sigma(t) = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}(\bar{t} - t_i)^2} \tag{12}$$

where N is the number of observations (50 in our case), $\bar{t}$ is the average time and t_i is the every time taken for each observation. With the value of s_t we are able to obtain $u_A(x)$.

$$u_A(t) = \frac{\sigma(t)}{\sqrt{N}} \tag{13}$$

On the other hand, $u_B(x)$, associated to the measuring instrument error is:

$$u_B(x) = \frac{\delta_x}{\sqrt{12}} \tag{14}$$

where δ_x is the resolution of the measuring instrument (δ_t=0,01s, δ_l=δ_D=0,001m and δ_m).

Finally, $u_C(x)$ is expressed as:

$$u_C(x) = \sqrt{u_A^2(x) + u_B^2(x)} \tag{15}$$

So, for t, we get:

$$u_C(t) = \sqrt{u_A^2(t) + u_B^2(t)} \tag{16}$$

5.2.2 Uncertainty of average speed

Referring to equation 7 and taking into account the above uncertainties, the uncertainty propagation can be calculated as:

$$u_C(v_m) = \sqrt{\left(\frac{\partial v_m}{\partial l}\right)^2 (u_C(l))^2 + \left(\frac{\partial v_m}{\partial t}\right)^2 (u_C(t))^2} = \sqrt{\left(\frac{1}{t}\right)^2 (u_C(v_m))^2 + \left(\frac{l}{t^2}\right)^2 (u_C(t))^2} \tag{17}$$

5.2.3 Uncertainty of limit speed

For the limit speed, v_l, obtained from equation 3, the uncertainty we get is:

$$u_C(v_l) = \sqrt{\left(\frac{\partial v_l}{\partial v_m}\right)^2 (u_C(v_m))^2 + \left(\frac{\partial v_l}{\partial D}\right)^2 (u_C(D))^2 + \left(\frac{\partial v_l}{\partial \phi}\right)^2 (u_c(\phi))^2} \tag{18}$$

5.2.4 Uncertainty of ball density

The uncertainty of this indirect magnitude whose value is obtained by dividing the mass of these by their volume is:

$$u_C(\delta) = \sqrt{\left(\frac{\partial \delta}{\partial m}\right)^2 (u_C(m))^2 + \left(\frac{\partial \delta}{\partial \frac{D}{2}}\right)^2 (u_C\left(\frac{D}{2}\right))^2} = \sqrt{\left(\frac{1}{\frac{4}{3}\pi(\frac{D}{2})^3}\right)^2 (u_C(m))^2 + \left(\frac{-3m}{\frac{4}{3}\pi(\frac{D}{2})^4}\right)^2 (u_C\left(\frac{D}{2}\right))^2} \tag{19}$$

5.2.5 Uncertainty of viscosity coefficient

This indirect magnitude can be calculated with the equation 3. Its corresponding uncertainty:

$$u_C(\eta) = \sqrt{\left(\frac{\partial \eta}{\partial D}\right)^2 (u_C(D))^2 + \left(\frac{\partial \eta}{\partial v_l}\right)^2 (u_C(v_l))^2 + \left(\frac{\partial \eta}{\partial \delta}\right)^2 (u_C(\delta))^2} =$$

$$\sqrt{\left(\frac{2D(\delta-\rho)g}{18v_l}\right)^2 (u_C(D))^2 + \left(\frac{-D^2(\delta-\rho)g}{18v_l^2}\right)^2 (u_C(v_l))^2 + \left(\frac{-D^2 g}{18v_l}\right)^2 (u_C(\delta))^2} \tag{20}$$

5.2.6 Uncertainty of Reynolds number

This is an indirect magnitude whose value is obtained with equation 2

$$u_C(R) = \sqrt{\left(\frac{\partial R}{\partial v_l}\right)^2 (u_C(v_l))^2 + \left(\frac{\partial R}{\partial D}\right)^2 (u_C(D))^2 + \left(\frac{\partial R}{\partial \eta}\right)^2 (u_C(\eta))^2} =$$

$$\sqrt{\left(\frac{\rho D}{\eta}\right)^2 (u_C(v_l))^2 + \left(\frac{\rho v}{\eta}\right)^2 (u_C(D))^2 + \left(\frac{-\rho v_l D}{\eta^2}\right)^2 (u_C(\eta))^2} \tag{21}$$

5.2.7 Uncertainty of the drag coefficient

The last one uncertainty, the drag coefficient´s one, can be expressed as:

$$u_C(C_D) = \sqrt{\left(\frac{\partial C_D}{\partial D}\right)^2 (u_C(D))^2 + \left(\frac{\partial C_D}{\partial \delta}\right)^2 (u_C(\delta))^2 + \left(\frac{\partial C_D}{\partial v_l}\right)^2 (u_C(v_l))^2} =$$

$$\sqrt{\left(\frac{4g}{3v_l^2}\right)\left(\frac{\delta}{\rho}-1\right)^2 (u_C(D))^2 + \left(\frac{4Dg}{3v_l^2\rho}\right)(u_C(\delta))^2 + \left(\frac{4Dg}{3v_l^3}\right)\left(\frac{\delta}{\rho}-1\right)^2 (u_C(v_l))^2} \tag{22}$$

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